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To: ImpulsoGov

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Improving Health Outcomes in Brazil

Previne Brasil ties federal primary care funding to 7 performance indicators to improve prevention, reduce complications, and save public spending. Our analysis shows that indicator scores improved in the early periods but then stalled, while key outcomes such as low birth weight and chronic disease hospitalisations barely moved, revealing a persistent gap between measured service delivery and real health gains.

We organise the work around three constraints. First, using a municipality-level event study around the 2024 elections, we find **clear drops in prenatal, diabetes, and hypertension KPIs after changes in mayor or party**, with flat pre-trends. This pattern is consistent with the replacement of community health workers, broken follow-up, and care cycles that must restart after each political turnover. Second, quadrimester panel fixed-effects analyses show that simply adding primary care teams is associated with only a slight reduction in chronic disease hospitalisations. In contrast, specialised maternal and oncology establishments and NASF teams are strongly associated with better neonatal outcomes. **Outcomes improve where care pathways are coordinated, not where visit counts alone rise.** Third, we document structural digital inequalities. A higher share of primary care teams without internet access is associated with higher diabetes and hypertension hospitalisations and weaker performance on coordination-intensive KPIs, indicating that **many municipalities cannot operate the digital backbone required by a performance-based model**.

These findings motivate three linked strategies rather than indicator redesign. Strategy 1, ***Cuidado Acima da Política***, protects continuity during political transitions through a 90-day period in which core teams remain in place, a structured handover of nominal lists and pending cases, and a simple knowledge management KPI, backed by public commitments between outgoing and incoming administrations. Strategy 2, ***Cuidado que Acompanha***, strengthens care pathways by extending existing dashboards so municipalities can see where referrals, tests, and prescriptions stall, and by agreeing on a minimal set of fields that providers routinely update. Strategy 3, ***Infraestrutura Forte***, ***Saúde Forte***, raises the minimum digital capacity by securing stable connectivity and basic equipment in the weakest units, supported by clearer maintenance contracts and targeted pilots, such as tax-credit-financed upgrades. Together, these proposals address continuity, pathway integration, and digital infrastructure, enabling municipalities to convert performance-based financing into durable, more equitable improvements in health outcomes.

Key Findings

Introduction: Better performance in primary care indicators does not translate into better health outcomes.

Previne Brasil introduced a results-oriented financing model for primary care, with federal transfers tied to seven national key performance indices (KPIs). The introduction of this scheme aims to improve primary health care, prevent health complications and diseases, and reduce public spending on their aggravation. Yet, looking at the joint evolution of performance indices and health outcomes, we see a clear mismatch between the two rather than a strong positive association, and a stall in the improvement of performance indices.[1]

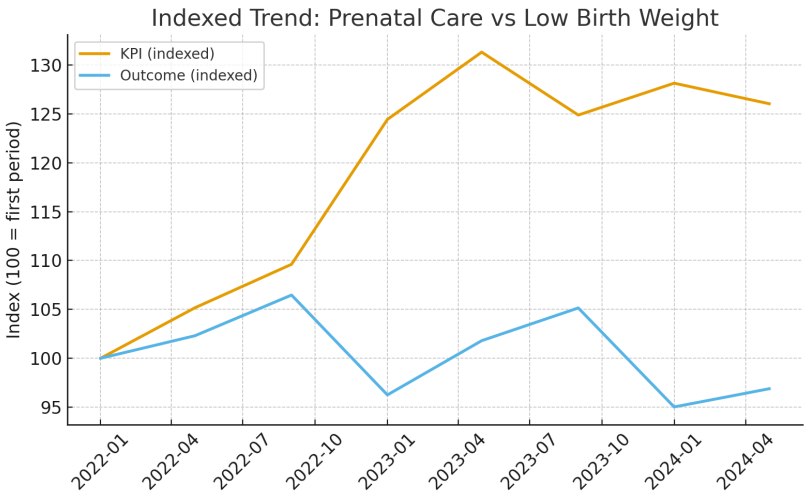


Figure 1. Indexed national trends in the prenatal care Key Performance Indicators and the share of low birth weight births, 2022–2024 (index 100 equals the first quadrimester).

The divergence between health outcomes and indicators suggests that the measured gains in service delivery inputs do not translate into proportional gains in health outcomes. While the stall suggests a problem in the continuation of the indicator tracking. Our working hypothesis is that three sets of constraints hinder the care pathway of Previne Brasil:

1. **Governance Failure:** institutional discontinuity around municipal elections disrupts the routines that sustain long-term follow-up, leading to sharp drops in KPI performance.
2. **Coordination in Referral to a Specialist:** even when KPIs improve, coordination across the care pathway remains weak. Referrals, lab results, imaging, and specialist care are not reliably linked back to primary care, so recorded contacts do not guarantee that the clinical problem is resolved.
3. **Structural Inequalities:** infrastructure and resource gaps limit what front-line teams can actually deliver once patients are identified.

In other words, primary care inputs stopped improving while outcomes stagnated, which motivates our focus on institutional stability, care coordination, and infrastructure as the likely binding constraints behind this mismatch and the main areas for improving Previne Brasil.

[1] Analysis of association between health outcomes and performance indicators, as well analysis of the stall progress, are in the appendix A2 Introduction section.

Problem 1: Institutional Discontinuity and Workforce Turnover as Symptoms of Governance Failure

Previne Brasil is contingent on government policies and critically dependent on local-level shifting political dynamics. Local experts expressed that in each municipal election, changes in political leadership unsettle the basic routines that sustain primary care. Rather than building on what the previous team achieved, many municipalities restart from scratch by replacing the entire health staff.

This institutional discontinuity weakens the system's capacity to follow patients over time, especially for chronic conditions and prenatal care. The local expert noted that health in Brazil is primarily driven by interactions between community workers and community residents. The community worker serves as a link between the health sector and patients, monitoring residents' well-being.

Thus, we hypothesize that *a shift in political dynamics, in this case, a political party change in the election on the municipal level, would lead to a drop in Previne Brasil's KPIs.* To formalize this pattern, we estimate a difference-in-differences event study in which the outcome is each performance indicator and the key regressors are time dummies relative to the election in municipalities with a party or mayor change.[2] We seek to identify the average estimated effect of political shifts on primary care indicators.

In the graph of prenatal visits, we see a **negative shock when a new party takes office**. As the community worker system is a key component of Brazil's healthcare, the replacement of these workers indicates that the professional experience unique to that region, let alone the trust with families, requires a lengthy rebuilding period.

Prevention and chronic disease management rely on repeated contacts over time, not single visits. Every disruption restarts the care cycle. Pregnant women or patients with chronic conditions lose follow-up, referrals fall through gaps, and treatment plans stall. Municipalities that had been steadily improving lose momentum and often need several quarters to return to their prior KPI levels, slowing cumulative gains.

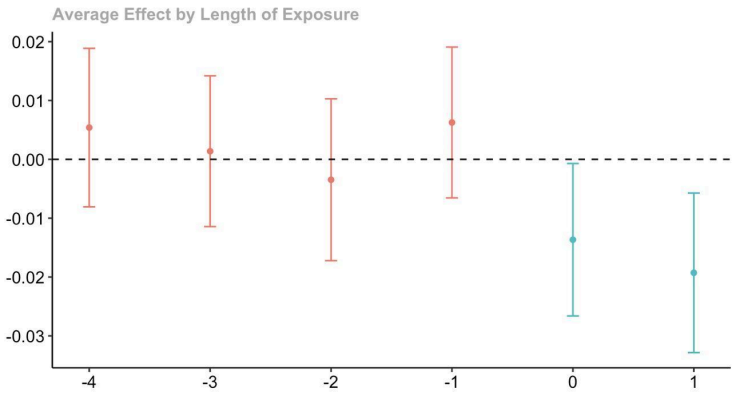


Figure 2. Difference in differences (DiD) estimates of the proportion of pregnant women with at least six prenatal visits, by time relative to municipal party change.

However, changes in the political landscape are inevitable in all democracies, as a NEJM article by Page and Polk (2017) examining the U.S. post-election health care shift found that, even in a country with rich administrative data, they cannot be avoided. In our Brazil case, the lack of data could only point to the actual effect being greater than the limited result we captured. Alleviating the turbulence on an institutional level, like training programs that are administered at a national level while providing an adequate buffer, ensuring a smoother transition, will be the solution that we look towards.

[2] Consult appendix A3 for the estimation in all key performance indicators by party or mayor change.

Problem 2: Weak Care Pathways: KPIs Measure Inputs, but Outcomes Depend on What Happens Next

After identifying problems through the institutional lens, we now move to further theorization to explain the gap between KPIs and outcomes, or between the primary care indicator and the specialized care outcome.

We hypothesize that *the link between citizens and the specialized health sector is constrained by poor coordination.* The health specialist shared that Brazil's primary care and specialized care databases are not integrated into a single information system. Primary care workers don't know whether the referral had follow-up if they don't have a support team called NASF (Núcleo de Apoio à Saúde da Família, modalidade).

NASFs are multi-professional teams created to support Family Health Strategy (ESF) primary-care teams. They usually include psychologists, nutritionists, physiotherapists, and social workers. They are formally linked to one or more ESF teams in a municipality. They primarily interact with residents through shared visits and community activities, while coordinating their work inside primary-care units and the municipal health network. In the absence of these teams, the follow-up to specialized care is more complicated, even with adequate monitoring of Previne Brasil's indicators.

To go beyond describing patterns and say something about possible effects, we estimate how outcomes change when a municipality has at least one NASF team, comparing municipalities with the same number of primary-care and specialised teams:[3]

- For chronic disease outcomes (hypertension and diabetes hospitalisations), once we make this comparison, the estimated effects of the number of primary-care teams and specialised clinics are very small and indistinguishable from zero, so we do not see clear evidence that simply having more teams changes these hospitalisations.

- For diabetes hospitalisations, the only clear association is with NASF presence: municipalities with NASF tend to have slightly higher diabetes hospitalisation rates. This is more consistent with NASF teams being placed in, and detecting more cases in, worse-off municipalities than with a simple "NASF prevents hospitalisations" story.

- For neonatal and maternal mortality, the picture is different: having more specialised maternal facilities is linked to lower mortality, and the presence of NASF teams is associated with an additional reduction in neonatal deaths.

Taken together, these results indicate **diminishing marginal returns to expanding primary care teams in isolation and point to where the data do show traction.** Outcome improvements align statistically with places that invest in specialised maternity and oncology infrastructure and that have NASF or similar multi-professional teams connecting services, rather than with simple increases in primary care capacity. The binding constraint is the organisation of the pathway that links primary and specialised care, not the volume of initial contacts that current KPIs record.

Problem 3: Structural Digital Inequalities Limit What Municipalities Can Achieve

Performance in Previne Brasil depends not only on how teams deliver care in clinics, but also on whether municipalities can operate the digital backbone of a performance-based system. We hypothesize that *weak digital infrastructure prevents municipalities from turning Previne Brasil incentives into better outcomes.*

Submitting nominal records, tracking indicators each quadrimester, and using data to guide outreach all require stable electricity, internet access, and basic equipment in health units and with community health agents. Local

[3] See appendix A4 for the model specification and coefficient results.

experts reiterated that in many municipalities, especially in the North and Northeast, these conditions are fragile or missing, so data entry is delayed, systems crash, and managers lack timely information on who is overdue for prenatal care, vaccines, or chronic disease follow-up.

To test this hypothesis, we follow each municipality every four months and track how its diabetes and hypertension hospitalisation rates, together with the Previne Brasil KPIs, change over time.[4] Our primary explanatory variable is digital infrastructure, measured as the proportion of primary care facilities in the municipality that report having internet access, built from establishment-level records and aggregated into a municipal index. To ensure we are not confusing digital gaps with simple capacity differences, we also control for the size of primary care and specialised teams, using the logarithm of the number of workers to ensure percentage changes in staff are comparable across both small and large municipalities.

The econometric specification uses fixed effects, meaning that we compare each municipality mainly to itself over time, net of its stable characteristics (such as geography, historical poverty, or baseline health profile) and of national shocks in each quarter. We also use lagged controls, so that past internet coverage and staffing explain current performance, reducing the risk that poor results mechanically trigger investments and bias the estimates.

Using this framework, we find statistically strong and substantively coherent evidence on where internet access matters. **A larger share of primary care teams <u>without internet</u> in the previous quarter is associated with <u>higher diabetes hospitalisation rates</u>**, with a large and highly significant coefficient, and with worse performance on coordination-intensive process indicators, in particular early **prenatal care and Pap smear coverage, where the lack of internet significantly <u>lowers KPI scores</u>**. These patterns align with the hypothesis that the internet is a systems constraint for tasks that require continuous tracking of patients, lab results, and referrals.

In contrast, coefficients for neonatal mortality and oncology-related procedures are small and not statistically different from zero, suggesting that other bottlenecks dominate those pathways, such as hospital quality and physical access. For several remaining KPIs, the estimated effect of poor internet is positive, which is counter to the clinical story and likely reflects indicator construction and denominator issues in low-connectivity settings rather than real performance gains.

Overall, the statistical results support a focused interpretation: internet connectivity is not a universal binding constraint, but it is highly relevant for chronic disease management, especially diabetes, and for selected maternal care processes that depend on reliable digital follow-up.

Recommendations and Implementation Plan

This section presents three strategies that are technically correct, politically supportable, and administratively feasible. They target the institutional, managerial, and digital constraints identified through our statistical analysis and interviews, and focus on mechanisms the Ministry and ImpulsoGov can realistically influence. The proposals below focus on the mechanisms that block progress and on actions that fall within the reach of the Ministry and ImpulsoGov.

Strategy 1. Protecting continuity during transitions "Cuidado Acima da Política"

Performance falls after elections because routines break, and new managers start without the operational knowledge needed to manage cases. Municipalities do not preserve this knowledge, which makes the first months of each administration costly. We suggest implementing a **<u>90-day continuity period in which the core primary care team and community health workers remain in place</u>**. The transition begins with a structured

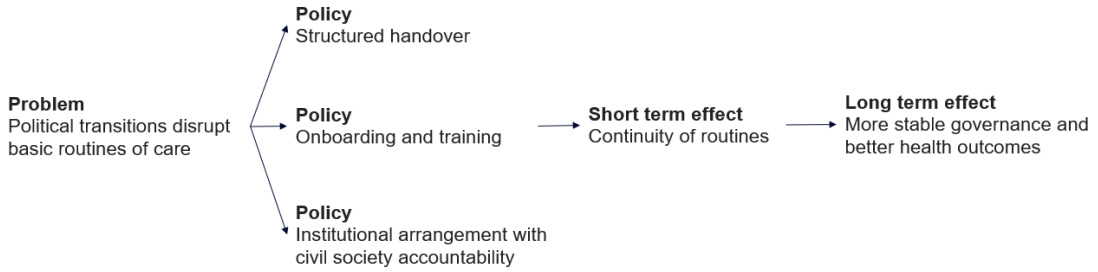
[4] See appendix A5. for the model specification and coefficient results.

handover. Outgoing and incoming authorities review updated nominal lists, micro-areas, active prenatal and chronic cases, pending referrals, and access to software and systems (PEC and e-SUS APS). ImpulsoGov can help organize this information so it is easy to use.

Then, ImpulsoGov and the outgoing technical team explain how indicators are built and which tasks require urgent attention in a three-day onboarding workshop. Priorities include confirming community health workers and their areas and reconciling pending production data. A simple knowledge-management KPI, monitored by ImpulsoGov, can track whether these routines are preserved during transitions. A civil society agreement is key to reinforcing the policy. A legal commitment is unlikely to advance now, but a public agreement between civil society and both administrations is feasible. The incoming team commits to keeping community health workers and core technical staff for 90 days, and an independent group reports on compliance with mass media using ImpulsoGov data, similar to the model used by "Argentinos por la Educación" in Argentina to monitor education.

Stakeholder assessment: Community health workers, technical teams, and most municipal managers benefit from continuity. Political actors may resist limits on early staffing decisions.
Capacity requirements: Municipalities need the capacity to maintain routine data and support community health workers. The Ministry, with ImpulsoGov, should provide training materials, ensure system access, and confirm that the handover is completed.

Figure 4. Theory of Change of "Cuidado Acima da Política"



Source: Own elaboration

Strategy 2. Make the patient visible "Cuidado que Acompanha"

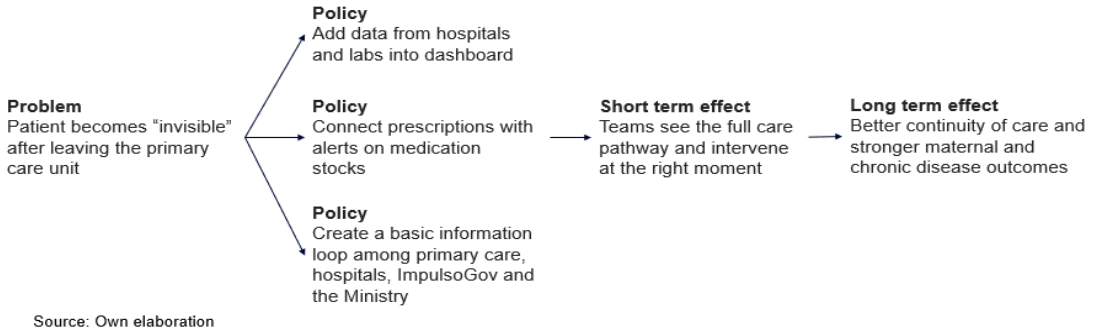
Municipalities with similar inputs perform differently depending on whether they have a clear link to specialised services or an NASF team for coordination. The bottleneck is the lack of timely information once a referral leaves the primary care. The proposal is to **extend the dashboard ImpulsoGov already provides.** This is the most feasible of the three strategies because it builds on a tool that municipalities use daily. In the short term, ImpulsoGov can pull the basic confirmations that hospitals and laboratories already record. Knowing whether a referral was received or whether the test or procedure was completed is enough to show where cases get stuck.

In the medium term, ImpulsoGov, municipalities, and the Ministry can agree on a small set of fields that each provider updates routinely. With this structure, the dashboard can also link prescriptions to basic information on medicine availability. Many units prescribe without knowing if the treatment is accessible. A simple alert on shortages for key prenatal and chronic medications would prevent silent treatment failures. The experience of the SALOG health PPP in Peru shows that when responsibilities are clear, simple digital routines can keep stock information reliable for frontline teams.

By exposing the missing information at the exact points where coordination breaks, municipalities can intervene earlier, reduce avoidable delays, and convert initial contacts into real improvements in maternal and chronic disease outcomes

Stakeholder assessment: Primary care teams and municipal managers benefit from clearer information. Some specialised services may hesitate if they expect extra work.
Capacity requirements: Municipalities need one person to update referral status and confirm medicine availability. ImpulsoGov adjusts the dashboards, trains teams and supports early use.

Figure 5. Theory of Change of "Cuidado que Acompanha"



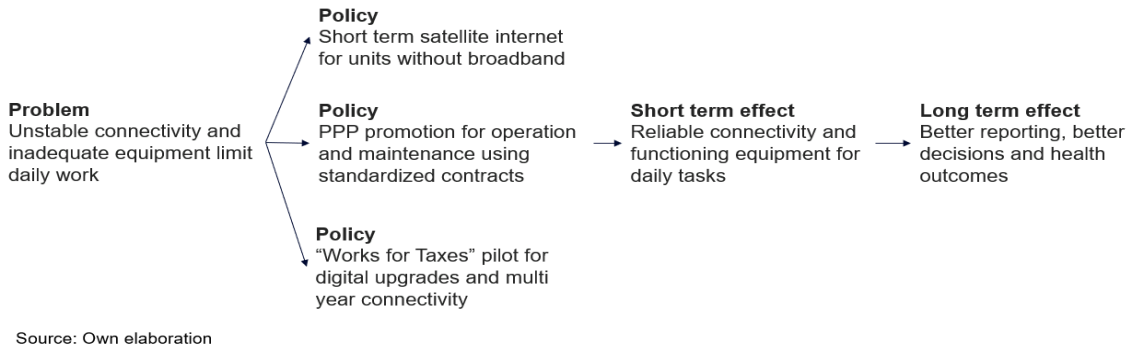
Strategy 3. Raising the minimum digital capacity "Infraestrutura Forte, Saúde Forte"

Units with unstable internet or inadequate equipment cannot properly follow cases. Our regressions show that municipalities with poor internet connectivity perform worse on coordination-intensive indicators. In the short term, we propose **providing stable connectivity to the weakest units**. Satellite internet, already used in remote schools, works well in isolated areas and can be contracted at the state or federal level. ImpulsoGov and municipalities then check whether the connection actually supports daily routines. Some municipalities need broader support. Partnerships focused on operation and maintenance are an option where budgets cannot cover repairs. Clearer contracts inspired by FIDIC or NEC give operators predictable rules and link payment to simple service standards that keep electricity, connectivity, and basic equipment functioning.

A small pilot can help the most remote municipalities. Following Peru's "Works for Taxes" model, companies finance essential upgrades or multi-year satellite internet and deduct the investment from their taxes. To target these efforts, ImpulsoGov should build a short index that combines internet stability with equipment conditions to identify municipalities with the most significant gaps and demand.

Stakeholder assessment: Technical teams welcome better conditions. Some municipalities may hesitate because of maintenance responsibilities.
Capacity requirements: Municipalities need the basic ability to maintain equipment. The Ministry oversees contracts and verification, with ImpulsoGov supporting prioritisation and monitoring.

Figure 6. Theory of Change of "Infraestrutura Forte, Saúde Forte"



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Technical Appendix

A1. Additional Data and Information Sources

CNES (Cadastro Nacional de Estabelecimentos de Saúde): is Brazil's official national registry of health facilities maintained by the Ministry of Health; it records, at the establishment level and updated regularly, detailed information on each facility's location, legal nature, type (e.g. basic units, hospitals, specialty centers), available services, human resources, beds, and selected infrastructure characteristics (such as equipment and support services). This makes CNES essentially a census-style administrative database of the supply side of the health system, which can be used to construct indicators of primary care, specialized care, and basic infrastructure capacity over time at the municipal level.

R package ElectionsBR: is an R package that provides a unified interface to download, clean, and harmonize Brazilian electoral data from the Superior Electoral Court (TSE) and the CEPESP Data project, covering multiple elections (municipal and national) and offices (mayors, councilors, deputies, governors, president), as well as complementary information on voter profiles, candidates' characteristics, social media, and financial disclosures.

Semi-structured interviews with professionals of Brazil: we conducted two one-hour interviews with a health expert and a policy expert who work in ImpulsoGov to recover the local context and reality so we could assess the plausibility of our diagnosis hypothesis and implementation strategies. We thank them for their insights and kindness.

 Beatriz Patrício **Project Analyst**	 **Isabela dos Santos** **Project Analyst**
Worked on Previne Brasil-related projects since 2022. Coordinated the relationship with the municipalities and her project. **Interviewed:** 2025-11-19	Primary care doctor by training. She practiced as a health worker for 3 years in Brazil's South during Previne Brasil Implementation. **Interviewed:** 2025-11-24
Guiding questions:	
From your experience, what are the biggest operational barriers municipalities face in meeting performance targets (data, staffing, processes, incentives)? How do political factors (mayors, secretaries, electoral cycle, turnover) affect a municipality's ability to benefit from P4P programs?	What are the most common "failure modes" you see: cases where municipalities want to improve but still don't meet targets? From your experience, what would be the shortcomings of our implementation strategies?

A2. Introduction technical notes

KPIs and outcomes first glimpse (Figure 1), we constructed national time series for each quadrimester from 2022 to 2025 for both all performance indicators (for example, the share of women with six or more timely visits) and the related outcome (the share of births with low birth weight). We then convert both series into indices that equal 100 in the first period and plot them on the same axis. This normalization removes level

differences and focuses attention on relative change. The prenatal care index increases by roughly 25-30% over the window, while the low birth weight index fluctuates around its baseline, with almost no net improvement.

KPIs' association with outcomes: The estimating equation in these regressions is, for each outcome y_it (e.g. diabetes hospitalizations, neonatal deaths) in municipality i and quadrimester t:

$$y_{it} = \beta KPI_{i,t-1}^{relevant} + \mu_i + \tau_t + \varepsilon_{it}$$

We didn't find a statistically significant association between the KPIs and their outcomes in most cases.

Municipality fixed effects absorb all time-invariant differences across municipalities (baseline risk profile, geography), and time fixed effects t absorb shocks common to all municipalities in a given quarter (national policy changes, macro shocks). This is the standard two-way fixed-effects panel specification (see e.g., Wooldridge, 2010, *Econometric Analysis of Cross-Section and Panel Data*). We cluster standard errors by municipality because we expect serial correlation and heteroskedasticity within municipalities over time, and we have around 5,000 clusters, so the usual cluster-robust asymptotics for inference are plausible. Finally, we use lagged KPIs rather than current KPIs to better match the intended mechanism ("preventive care today, better outcomes tomorrow").

Inequality Among Municipalities. When averaging across all seven indicators, municipal inequality (P90 - P10 gap; summarizing the extent of municipal dispersion across all indicators) declined from about 52 percentage points in early 2022 to around 45 points in 2024, before stabilizing near 46-47 points in 2025. Even after this improvement, top-decile municipalities still outperform bottom-decile ones by roughly half the coverage scale, and the most unequal indicators continue to show gaps above 60 percentage points in several periods, while even the least unequal indicators rarely fall below 30 percentage points. This pattern indicates that convergence has been modest and that considerable performance differences between municipalities remain.

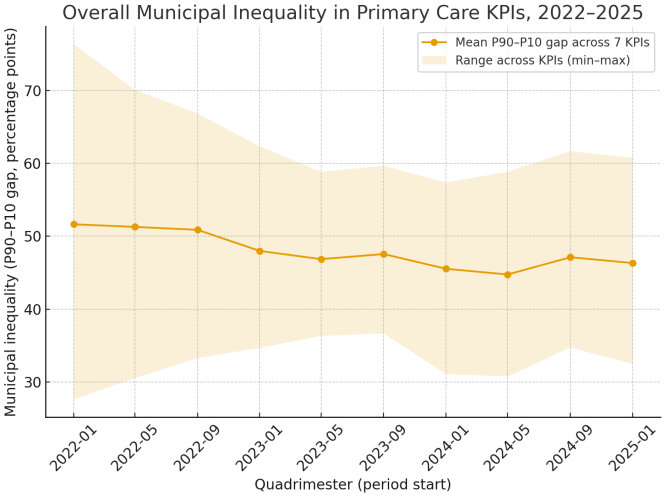

Figure A2.1 Overall municipal inequality in primary care KPIs, 2022–2025 (mean P90–P10 gap, with range across indicators).

A3. Problem 1. Institutional shift as a symptom of bad governance: technical notes.

This difference-in-differences specification uses Callaway and Sant'Anna's approach to compare how primary-care KPIs evolve in municipalities that experienced a change in mayor or party in the 2024 election versus municipalities that were never treated. We interpret the estimated ATT as the average change in the KPI after the political change, net of nationwide shocks and time-invariant municipal characteristics. Identification relies on two strong assumptions: i) that trends in the absence of treatment would have been parallel, which looks reasonable given the flat and non-significant pre-treatment coefficients; and ii) that the timing of

mayor/party changes is “as good as random” with respect to future KPI trends, which is a much weaker assumption here because electoral outcomes are not random. Under these caveats, the dynamic DiD estimates show that most KPIs exhibit a negative, statistically significant overall ATT after a major or party change, suggesting that, on average, preventive primary-care performance deteriorates following the 2024 electoral turnover.

Table 1. Difference in Difference Overall Average treatment on the treated estimation

Indicator	Treatment	ATT	Std. Err	CI low	CI high	95%
Proportion of 1-year-old children vaccinated in primary care against diphtheria, tetanus, pertussis, hepatitis B, Haemophilus influenzae type b, IPV	Mayor change	-0.0015	0.0034	-0.0080	0.0051	
Proportion of 1-year-old children vaccinated in primary care against diphtheria, tetanus, pertussis, hepatitis B, Haemophilus influenzae type b, IPV	Party change	0.0001	0.0037	-0.0072	0.0074	
Proportion of people with diabetes who had a visit and an A1C test ordered in the 6-month period	Mayor change	-0.0153	0.0035	-0.0221	-0.0085	*
Proportion of people with diabetes who had a visit and an A1C test ordered in the 6-month period	Party change	-0.0101	0.0036	-0.0171	-0.0030	*
Proportion of people with hypertension who had a visit and blood pressure measured in the 6-month period	Mayor change	-0.0119	0.0027	-0.0171	-0.0066	*
Proportion of people with hypertension who had a visit and blood pressure measured in the 6-month period	Party change	-0.0056	0.0027	-0.0109	-0.0003	*
Proportion of pregnant women tested for syphilis and HIV	Mayor change	-0.0127	0.0047	-0.0219	-0.0035	*
Proportion of pregnant women tested for syphilis and HIV	Party change	-0.0086	0.0044	-0.0173	0.0000	*
Proportion of pregnant women who received dental care	Mayor change	-0.0090	0.0046	-0.0180	0.0000	
Proportion of pregnant women who received dental care	Party change	-0.0117	0.0049	-0.0213	-0.0021	*
Proportion of pregnant women with at least 6 prenatal visits, with the first by the 12th week of gestation	Mayor change	-0.0082	0.0043	-0.0167	0.0003	
Proportion of pregnant women with at least 6 prenatal visits, with the first by the 12th week of gestation	Party change	-0.0165	0.0046	-0.0254	-0.0075	*
Proportion of women with cervical cytology (Pap smear) collected in primary care	Mayor change	-0.0027	0.0009	-0.0046	-0.0009	*
Proportion of women with cervical cytology (Pap smear) collected in primary care	Party change	-0.0016	0.0011	-0.0036	0.0005	

A4. Problem 2. KPIs Measure Inputs, but Outcomes Depend on What Happens Next

We estimate, for each outcome y_it, the following municipality–quadrimester fixed-effects model, weighted by the relevant population denominator:

$$y_{iq} = \beta_1 \, \mathrm{PCteams}_{i,q-1} + \beta_2 \, \mathrm{SpecEstab}_{i,q-1} + \beta_3 \, \mathrm{NASF}_{i,q-1} + \mu_i + \lambda_q + \varepsilon_{iq},$$

Where PCteamsi,t−1 is the number of primary-care teams in municipality i in the previous quadrimester, SpecEstabi,t−1 is the number of condition-specific specialised establishments, NASFi,t−1 is an indicator equal to 1 if the municipality had at least one NASF team. β3 captures the within-municipality association between having NASF and the outcome.

Table 2. Fixed effects regressions showing how primary care teams, specialised services, and NASF are associated with screening, hospitalisations, and neonatal deaths across municipalities.

Outcome	Lagged structural variable	Estimate	Std. error	p value	Significance
Outpatient cytology procedures per target population	Oncology establishments (n_estab_onco_l1)	4.50×10^{-7}	1.58×10^{-7}	0.0044	Yes (1%)
	Primary care teams (n_pc_teams_l1)	-8.09×10^{-8}	2.25×10^{-8}	0.00034	Yes (1%)
	Oncology reference centre (has_onco_ref_l1)	-1.41×10^{-5}	1.55×10^{-5}	0.36	No
Hospitalisations due to hypertension per HTN population	Primary care teams (n_pc_teams_l1)	-4.67×10^{-9}	1.00×10^{-8}	0.64	No
	Chronic NCD establishments (n_estab_chronic_ncd_l1)	-1.95×10^{-8}	1.57×10^{-8}	0.21	No
	NASF present (has_nasf_l1)	3.99×10^{-6}	3.43×10^{-6}	0.24	No
Hospitalisations due to diabetes per diabetes population	Primary care teams (n_pc_teams_l1)	3.44×10^{-9}	2.15×10^{-8}	0.87	No
	Chronic NCD establishments (n_estab_chronic_ncd_l1)	1.63×10^{-7}	1.14×10^{-7}	0.16	No
	NASF present (has_nasf_l1)	4.31×10^{-5}	1.54×10^{-5}	0.005	Yes (1%)
Neonatal deaths per births	Primary care teams (n_pc_teams_l1)	3.30×10^{-6}	4.70×10^{-7}	$<1\times10^{-11}$	Yes (1%)
	Maternal/neonatal establishments (n_estab_maternal_l1)	-2.04×10^{-5}	4.71×10^{-6}	$\approx1.4\times10^{-5}$	Yes (1%)
	NASF present (has_nasf_l1)	-8.65×10^{-4}	4.13×10^{-4}	≈0.036	Yes (5%)

All models: municipality by quadrimester panel, municipality and period fixed effects, weighted by the relevant population denominator.

A5. Problem 3. Structural Digital Inequalities Limit What Municipalities Can Achieve

$$y_{iq} = \beta_1 \,\text{share_no_internet}_{i,q-1} + \beta_2 \,\log\!\big(1 + \text{SpecEstab}/\text{PCteam}_{i,q-1}\big) + \mu_i + \lambda_q + \varepsilon_{iq},$$

share_no_internet_i_t−1 is the share of primary-care teams without internet in the previous quarter. We use the logarithm of the number of establishments/teams because these counts are highly skewed across municipalities. We are interested in proportional rather than absolute changes: the coefficients can be read as the change in the outcome associated with a given percentage increase in service capacity.

Outcome	Lagged structural variable	Estimate (β)	p value	Stat. sig.
Hospitalisations due to hypertension per HTN population	Share of PC teams without internet (share_no_internet_l1)	5.8×10^{-6}	≈ 0.047	Yes (5 percent)
	Chronic NCD establishments (log1p(n_estab_chronic_ncd_l1))	≈ 0	> 0.10	No
Hospitalisations due to diabetes per diabetes population	Share of PC teams without internet (share_no_internet_l1)	3.47×10^{-5}	$<1\times10^{-6}$	Yes (1 percent)

	Chronic NCD establishments (log1p(n_estab_chronic_ncd_l1))	≈ 0	> 0.10	No
Neonatal deaths per births	Share of PC teams without internet (share_no_internet_l1)	-1.7×10^{-4}	≈ 0.77	No
	Maternal and neonatal establishments (log1p(n_estab_maternal_l1))	≈ 0	> 0.10	No
Outpatient cytology procedures per target population	Share of PC teams without internet (share_no_internet_l1)	Small, < 0	> 0.10	No
	Oncology reference centre (has_onco_ref_l1)	≈ 0	> 0.10	No
KPI 1 – early and sufficient prenatal care (result_pct)	Share of PC teams without internet (share_no_internet_l1)	−0.054	~ 1×10^{-4}	Yes (1 percent)
KPI 4 – cervical cytology in primary care (result_pct)	Share of PC teams without internet (share_no_internet_l1)	−0.021	≈ 0.007	Yes (1 percent)
KPI 2, 3, 6, 7 – HIV/syphilis testing in pregnancy; dental care in pregnancy; HTN and DM follow up (result_pct)	Share of PC teams without internet (share_no_internet_l1)	> 0 (varies by KPI)	< 0.01	Yes (1 percent)
KPI 5 – childhood vaccination (result_pct)	Share of PC teams without internet (share_no_internet_l1)	≈ 0	> 0.10	No
All KPIs	Primary care teams (log1p(n_pc_teams_l1))	> 0 (most KPIs)	Often < 0.01	Yes for most

All models: municipality by quadrimester panel, municipality and period fixed effects, weighted by the relevant population denominator.

Stakeholder landscape

Stakeholder	Strategy 1	Strategy 2	Strategy 3
Municipal governments	Lead transition and handover	Provide staff to update referral status and coordinate with hospitals	Receive connectivity upgrades and maintain equipment
Community Health Workers	Develop territorial knowledge	Use dashboard to track patients	Benefit from stronger connectivity and report cases more reliably.
ImpulsoGov	Organizes handover, provides onboarding and monitors KPI	Expands dashboards and trains teams	Build digital index and monitors improvements
Ministry of Health	Verifies compliance	Endorses required fields and supports integration	Contracts satellite internet, oversees O&M and sets standards
Hospitals & laboratories		Provide referral and test confirmations.	
Civil society	Monitors' transition commitment		
Private partners			Provide connectivity, upgrades, O&M